

NCERT

National Cyber Emergency Response Team

DIGITAL FORENSICS LABORATORY — GROUP PROJECT REPORT

Evidence Management System (with Chain of Custody)

Phase 1 Report — Research, Requirements, and System Foundation
25% Project Completion

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Academic Integrity Notice: All design, implementation, and verification activities were performed on an isolated, instructor-authorized laboratory / development system for the NCERT Digital Forensics Internship. No production case systems or third-party operational evidence stores were accessed at any stage of this project.

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Abstract

This report documents **Phase 1 (25% completion)** of Group 2’s internship project *Evidence Management System (with Chain of Custody)*, undertaken under the **National Cyber Emergency Response Team (NCERT)** Digital Forensics Internship and supervised by **Sir Umar Javaid**. Phase 1 concentrates on research, requirements analysis, and foundational design rather than full module implementation. The team studied peer-reviewed and practitioner guidance on digital chain of custody, SHA-256 integrity verification, role-based access control (custodian, examiner, supervisor), and forensic audit trails; mapped these findings to the official project brief; and produced a concrete architecture, database schema, and implementation roadmap for subsequent phases.

1 Introduction

Digital evidence loses courtroom value when custody history is incomplete or when file integrity cannot be proven. Traditional paper logs do not scale for high-volume digital exhibits. Group 2’s assigned deliverable is a web-based Evidence Management System that registers uniquely identified digital items, computes SHA-256 hashes at ingest, records every custody event, re-hashes on handoff to detect tampering, and enforces role-based access with a full audit trail. Phase 1 establishes the research foundation and engineering blueprint required before coding the operational modules.

1.1 Objectives

- 1 Interpret the official NCERT project brief and define measurable success criteria.
- 2 Review literature and practice on digital chain of custody and hash-based integrity.
- 3 Elicit functional and non-functional requirements for all four brief scope items.
- 4 Select a modern, maintainable technology stack suitable for a forensic lab demo.
- 5 Design the logical architecture, RBAC matrix, and relational data model.
- 6 Produce a phased implementation plan covering the remaining 75% of the work.

1.2 Scope

Phase 1 is limited to research, requirements, architecture, and design artefacts. Working application modules (evidence registration UI, live custody ledger, integrity queue, PDF reports) are deferred to Phases 2 and 3. Physical evidence seizure and hardware imaging remain out of scope; the system manages *already acquired* digital exhibits and their custody metadata.

1.3 Tools and Environment

Component	Details
Host OS	Windows 10/11 development workstation
Primary stack (planned)	Next.js 14, TypeScript, Prisma, PostgreSQL
Authentication (planned)	NextAuth.js (Credentials + JWT)
Integrity algorithm	SHA-256 (Node.js crypto, server-side only)
Design artefacts	ER diagrams, RBAC matrix, module roadmap
Reference brief	Project 1 — Evidence Management System (with CoC)
Version control	Git repository (local project workspace)

Table 1: Investigation / development tools and environment (Phase 1).

2 Problem Statement and Project Brief

The official brief describes the system as “the flagship evidence-tracking platform — inventory meets custody ledger.” It must register digital evidence as uniquely identified items while maintaining a

tamper-evident chain-of-custody log, and double as an inventory of who held an exhibit, when, and why. The deliverable type is an **Evidence / inventory management system** within the forensic domain of evidence / case workflow.

#	Brief requirement	Phase-1 research focus
1	Register items with unique IDs, intake details, SHA-256 at ingest	ID schemes, hash-at-ingest patterns, intake metadata fields
2	Record seizure / transfer / examination / return with handler, time, location, reason	Custody event taxonomy, append-only ledger design
3	Re-hash on handoff; flag mismatches; printable custody report	Integrity workflows, report content for court packaging
4	RBAC (custodian, examiner, supervisor) + full audit trail	Permission matrices, audit append-only semantics

Table 2: Official project scope mapped to Phase-1 research questions.

3 Research and Literature Review

3.1 Chain of Custody in Digital Evidence

Chain of custody (CoC) is the chronological documentation of seizure, custody, control, transfer, analysis, and disposition of evidence. In digital forensics, CoC must also prove that the bits examined are the bits seized. Carrier's foundational treatment of file-system forensics emphasises that documentation and integrity hashing are inseparable from technical acquisition [1]. Practitioners therefore treat every movement of an exhibit as an auditable event with actor, timestamp, location, and stated reason.

Paper ledgers remain common in some labs but are error-prone and hard to search. Electronic CoC systems reduce omission risk when they enforce mandatory fields, bind events to authenticated users, and prevent silent rewrites of historical records. Phase 1 therefore adopts an **append-only custody ledger** as a first-class design constraint.

3.2 Cryptographic Integrity (SHA-256)

Cryptographic hashing provides a compact fingerprint of exhibit bytes. SHA-256 is widely accepted in forensic practice for imaging and file verification [2]. Computing the digest at ingest establishes a baseline; recomputing it on each handoff detects accidental corruption or intentional tampering. A mismatch must never be silently overwritten: the correct operational response is to flag the item, block further routine movement, and require supervisory resolution with a written note.

3.3 Role-Based Access Control in Forensic Labs

Forensic organisations separate duties so that no single actor can unilaterally alter evidence status without oversight. The project brief names three operational roles—**custodian**, **examiner**, and **supervisor**. Phase 1 extends this with an **admin** role for user lifecycle management, consistent with least-privilege practice. Permissions must be enforced both at the route layer and inside every server-side mutation, not only in the user interface [3].

3.4 Relevance to Pakistani Cyber-Forensic Practice

Haque et al. document gaps in Pakistan's cyber-forensic investigation readiness spanning policy, skill, organisation, and tooling [4]. A lightweight, open-stack evidence inventory with cryptographic CoC directly supports training objectives at NCERT: trainees practise intake hashing, documented handoffs, integrity flags, and printable reports without depending on expensive commercial case-management suites. Group 2's system is therefore both a software deliverable and a pedagogical platform.

4 Requirements Analysis

ID	Requirement	Priority
FR-01	Unique evidence IDs (e.g., EVD-YYYY-NNNN) at registration	Must
FR-02	Capture intake metadata (case ref, type, location, notes)	Must
FR-03	Compute/store SHA-256 at ingest (upload or double-entry hash)	Must
FR-04	Custody events: SEIZURE, TRANSFER, EXAMINATION, RETURN	Must
FR-05	Record handler, timestamp, location, reason on every event	Must
FR-06	Re-hash on handoff; flag INTEGRITY_FLAGGED on mismatch	Must
FR-07	Supervisor resolution workflow for flagged items	Must
FR-08	Printable / downloadable custody report (PDF) per item	Must
FR-09	RBAC for custodian, examiner, supervisor (+ admin)	Must
FR-10	Append-only system audit trail with export	Must
NFR-01	Server-side permission checks on all mutations	Must
NFR-02	Responsive UI suitable for lab demo / internship review	Should

Table 3: Consolidated functional requirements (elicited in Phase 1).

5 System Architecture and Technology Selection

After comparing monolithic desktop tools with a web application, Group 2 selected a **Next.js App Router** architecture: server components for secure data access, server actions for mutations, and PostgreSQL via Prisma for durable storage. This stack supports typed models, role-aware middleware, and PDF generation while remaining deployable for demonstration.

Layer	Choice	Rationale
UI / App	Next.js 14 + TypeScript + Tailwind	SSR security, typed routes, fast UI
Auth	NextAuth Credentials + JWT	Role claims in session; lab-friendly demos
Database	PostgreSQL + Prisma	Relational integrity; migrations; seed data
Hashing	Node crypto SHA-256	Server-only; no client-trusted digests
Reports	@react-pdf/renderer	Printable custody packages
Charts	Recharts	Dashboard custody / integrity overview

Table 4: Technology decisions and rationale.

6 Data Model and Security Design

The logical model centres on **User**, **EvidenceItem**, **CustodyEvent**, **IntegrityCheck**, and **AuditLogEntry**. Evidence items store the baseline SHA-256 and current custodian; custody events form an ordered ledger; integrity checks record recomputed digests and mismatch outcomes; audit entries capture security-relevant actions without update/delete APIs.

Capability	Custodian	Examiner	Supervisor	Admin
View dashboard / evidence / custody	Yes	Yes	Yes	Yes
Register evidence	Limited	Yes	Yes	Yes
Log custody events	Yes	Yes	Yes	Yes
Integrity re-hash / review	No	Yes	Yes	Yes
Resolve integrity flags	No	No	Yes	Yes
Generate custody PDF	No	Yes	Yes	Yes
View full audit trail	No	No	Yes	Yes
Manage users / roles	No	No	No	Yes

Table 5: Draft RBAC permission matrix (design baseline for Phases 2–3).

7 Phase-1 Deliverables and Progress (25%)

#	Deliverable	Status
1	Project brief interpretation & problem statement	Completed
2	Literature / practice review (CoC, hashing, RBAC)	Completed
3	Functional & non-functional requirements (Table 3)	Completed
4	Technology stack selection with rationale	Completed
5	High-level architecture & module boundaries	Completed
6	Draft ER / data model & RBAC matrix	Completed
7	Phased implementation roadmap (25 / 50 / 100%)	Completed
8	Runnable evidence / custody UI modules	Deferred → Phase 2
9	Integrity, reports, audit, admin completion	Deferred → Phase 3

Table 6: Phase 1 completion checklist (25% of total project).

8 Results Summary

Phase 1 achieved its intended 25% milestone: the team can articulate exactly what the system must do, why each brief requirement matters forensically, which stack will implement it, and how roles and hashes will be enforced. No production evidence was processed; all work was design- and research-oriented on authorised internship resources.

Expectation	Outcome	Evidence
Clear problem framing	Met	§2; Table 2
Research depth	Met	§3; References
Requirements baseline	Met	Table 3
Architecture readiness	Met	§5–§6; Tables 4–5
Honest progress claim (25%)	Met	Table 6

Table 7: Phase 1 results against internship expectations.

9 Conclusion

Phase 1 established a defensible foundation for the Evidence Management System with Chain of Custody. Research confirmed that unique IDs, ingest-time SHA-256, append-only custody events, handoff re-hashing, RBAC, and audit logging are the minimum set needed to satisfy the NCERT brief. With architecture and requirements locked, Group 2 proceeds to Phase 2 to implement authentication, evidence registration, and the custody ledger—the first operational half of the platform.

References

- [1] B. Carrier, *File System Forensic Analysis*. Addison-Wesley.
- [2] NIST, "Secure Hash Standard (SHS)," FIPS PUB 180-4.
- [3] R. S. Sandhu et al., "Role-based access control models," *IEEE Computer*, 1996.
- [4] E. U. Haque et al., "Cyber forensic investigation infrastructure of Pakistan," *IEEE Access*, vol. 11, 2023.
- [5] NCERT Digital Forensics Internship — Project 1 brief: Evidence Management System (with Chain of Custody).
- [6] Group 2 Week 3 laboratory report style reference — File-System Forensics and Data Recovery (NCERT format).
- [7] SWGDE, "Best Practices for Computer Forensics," current revision.
- [8] ISO/IEC 27037, Guidelines for identification, collection, acquisition and preservation of digital evidence.

A Phase Plan Overview

Phase	Weight	Focus
Phase 1 (this report)	25%	Research, requirements, architecture, data/RBAC design
Phase 2	25% (→ 50%)	Auth, evidence ingest + SHA-256, custody events, dashboard core
Phase 3	50% (→ 100%)	Integrity, PDF reports, audit, admin, polish, verification

Table 8: Three-phase completion plan for the EMS internship project.